University of Groningen

Improving Machine-learning Classifiers for Low Redshift Emission-line Galaxies

Final report for *Deep Learning in Physics*

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1 Introduction

Galaxy classification has been an important goal of astronomy for over a century, most famously, Hubble (1926) devised a classification scheme for galaxies based on their morphology which forms the foundation of our current classification. Galaxies can be classified on other characteristics apart from morphology, for example, based on the dominant source of their ionized gas emission. Ionized regions can tell us much about dominant mechanisms such as active galactic nuclei (AGN), strong star formation, metallicity and chemical evolution in galaxies. In addition, they can be observed with optical and infrared (IR) telescopes which are some of the most common telescopes since these wavelengths are easier to observe than X-ray (these require space-based telescopes) and radio waves (which require large arrays). This type of classification splits galaxies into four types: star-forming galaxies, where the main source of ionization is recent/ongoing star formation; AGN, where the main source of ionization is a (supermassive) black hole accreting material; composite galaxies in which the ionization source is a mix between star formation and AGN; and low-ionization nuclear emission regions (LINERs), where weakly ionized atoms dominate.

Galaxy classification into these four categories is important to ensure further analysis of the galaxies is done correctly. Different classes have different underlying physical processes, different excitation and ionization conditions, and thus if we want to apply an analysis technique that is calibrated to one type of galaxy, it could give incorrect results on other types of galaxies. Thus being able to classify galaxies in subtypes just from their emission line observations without too many underlying assumptions is key for further analysis of these galaxies. Additionally, understanding the population of different types of galaxies at higher redshifts tells us how galaxies evolve over cosmic time.

This classification is the main goal of the paper that we chose for our report: Zhang et al. (2019). They use machine learning to classify galaxies into one of these four categories. They train their models on low-redshift data and use them to classify intermediate redshift data. Machine learning can be a powerful tool to classify galaxies, morphological classification from images has been widely implemented using machine learning (e.g. Willett et al. (2013); Dieleman et al. (2015)). However, machine learning can also be a powerful tool to classify galaxies using spectral data, mainly using chemical line strengths in the spectra (e.g. Cavuoti et al. (2014)), which is the method that Zhang et al. (2019) uses.

We start our report with a summary of the paper, some reflections on it, and introduce the data used. In the next section, we detail how we reproduced the paper's results, how we built our own improved models, and present our results. Finally, we discuss our results by comparing them to the paper and conclude our project with general considerations and indications for future work.[1]

[1]The link to our GitHub repository that has all our code and results: https://github.com/lorlo9999/DeepLearningProject

2 The paper: 'Machine Learning Classifiers For Intermediate Redshift Emission Line Galaxies'

2.1 Paper summary

The aim of Zhang et al. (2019) is to classify intermediate redshift ($z = 0.32 - 0.8$) optical emission line galaxies into four different categories, star-forming galaxies, composite galaxies, AGN, and LINERs. To achieve this, the [N II], Hα, and [S II] lines are needed. However, these lines are redshifted out of the optical spectrum range in which these galaxies are observed. Thus, they train four different machine-learning models using lower redshifted ($0 - 0.32$) galaxies, which they then apply to classify the higher redshift galaxies.

Low redshift galaxies can be classified using optical diagnostic diagrams such as BPT diagrams (Baldwin et al., 1981) which classify galaxies based on relative line strengths of [O III]/Hβ, [N II]/Hα, [S II]/Hα, and [O I]/Hα. in the paper, they specifically chose to use BPT diagrams since other diagnostic diagrams usually struggle to separate composite galaxies and LINERs from the other two categories, AGNs and SFGs.

The data they use is from the SDSS-IV DR15 Extended Baryon Oscillation Spectroscopic Survey (eBOSS). Some selection criteria is used on the data, namely the signal-to-noise ratio of the line widths should be larger than 3 and the velocity dispersion should be larger than zero. This results in 28,869 low redshift galaxies, which are split into 70/30 training/validation samples. Additionally, they use 6-fold splitting. These galaxies are then classified using BPT diagrams (Kauffmann et al., 2003a; Kewley et al., 2006), which results in quite imbalanced classes. Due to this, they created a new sample with equally weighed classes by randomly choosing from each class. The intermediate redshift sample which they classify consists of 49,272 galaxies.

Four different machine learning algorithms are used and compared in the paper: k-nearest neighbors (KNN), support vector classifier (SVC), random forest (RF), and a multilayer perceptron neural network (MLP-NN). The eight input features for each of these are the following: the relative line strengths [O III]/Hβ, [O II]/Hβ, the line width $\sigma_{[\text{O III}]}$, the stellar velocity dispersion σ_*, and the colors $u - g$, $g - r$, $r - i$, and $i - z$. They chose these colors specifically because the SDSS survey provides these colors for about a third of the sky and they have a high signal-to-noise ratios. The other features are chosen because they can easily be measured from the optical spectra of the sources for galaxies with redshifts below 0.8, [O II], Hβ, and [O III] are the strongest lines below 5010 Å, and the stellar velocity dispersion can be well-measured using continuum fitting.

From the four different machine learning methods, the paper found that the random forest performed best, with the neural network as a close second. Their results are shown in Table 3 and Table 4. They compare the performance of the classifiers using per-class and balanced accuracy, ROC curves and the respective AUC scores.

Finally, they apply their trained RF model to the intermediate redshift data. The correctness of the classifications cannot yet be verified, but this will be possible once infrared spectra of the sources are available since the lines needed for the BPT classification are included in the infrared spectrum. This may be possible with new data such as that from Kriek et al. (2015) and Sanders et al. (2016), however as far as we can tell, no such data has yet been released for these sources.

2.2 Reflection on the paper

Our main criticism of the paper is that they determine the labels for the low redshift data themselves, which may cause some of the sources to be classified incorrectly. The BPT diagram diagnostic is a widely accepted classification method and the paper seems to exclusively use the classification scheme described in Kewley et al. (2006). This classification scheme is built up from such diagrams and considers several different lines to create these. However, in Zhang et al. (2019), for reasons that they do not discuss, they omit the second criteria for star-forming galaxies, the third for AGN, and the third and fourth for LINERs (Equations 2, 8, 12, and 14 in Kewley et al. (2006)). Three of these include [O I] lines, which their data included, but they did not use at all, the fourth one (Equation 12) includes the [S II] line which they used for other criteria too, so they could have easily used this criteria too. Omitting these criteria might have led to some incorrect classifications.

Another question that arose that is not addressed in the paper is having a fifth category, which would include sources that are in none of the other four categories, or where the category is unclear, such as the *"Ambiguous galaxies"* category in Kewley et al. (2006). This might be because they omitted the diagram featuring the [O I] line, so there is less overlap in the categories, but even with just the other two diagrams there is some overlap between the categories.

A second criticism is that there is no way to verify that the classification of the intermediate redshift data is correct. Once new, longer wavelength (IR) data of these specific galaxies is obtained, the classification could be assessed, however, as far as we could find, this data has either not been obtained or not been made publicly available. Therefore, there is no way to tell if these models will correctly predict the classes of the intermediate redshift galaxies that the paper classified since the features might not scale the same with redshift.

Another way to verify the classification would be to verify it with other data that has been classified (i.e. has IR observations) or can be classified. This leads us to the final criticism of the paper: their models do not generalize to other data sets. They mention this briefly in the first paragraph of Section 4. We examined whether the model generalizes to intermediate redshift galaxies, by obtaining data from the SDSS survey in the intermediate redshift range. However, neither our nor their models managed to classify the new data correctly, as we discuss in Section 4. The reasons could be different data pipelines resulting in slightly different features, but it could also mean that this approach for generalizing a model trained on low redshift does not work.

2.3 Data Size and Class Imbalance

According to the paper, the training set consists of 28869 galaxies, taken from a dataset of almost 60 thousand. Two issues came up regarding the data and the code almost instantly. These are that the distribution of classes among the dataset is unbalanced, and after adapting and running their code our 'working' dataset is much smaller than what they claim to have trained/tested their classifiers on. Even though we applied their exact way of filtering the data, we are now working with a dataset of *6533* sources.

The authors claim to balance out their classes in the paper before training, yet we did not find this process in the publicly available code. After running their filters on the available data we obtain a final distribution of classes in our training and test sets, which we show in Figure 1. This is clearly an unbalanced situation, where Star-Forming galaxies make up almost 60% of our dataset. We do not then balance the classes for two main reasons: firstly, we only have 6533 sources while they had almost 28869, and balancing out the classes would leave us with even less data. Secondly, we want to test if we can make models robust enough to compare to, or even outperform theirs, using a different approach to the data problem than class balancing.

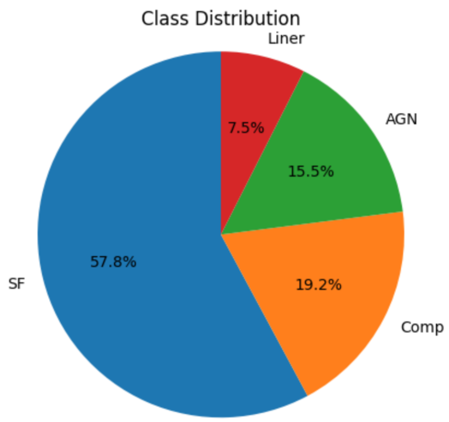


Figure 1: Pie chart of class distribution in our dataset

We therefore decided to use the method of threshold adjustment to deal with the class imbalance in our data. By manually adjusting the (probability) classifying thresholds of our models we can fine-tune them, so that it is easier for an object of one of the minority classes to be classified as such, while retaining a high balanced accuracy. Essentially, we manually define a probability threshold per class that overrides the classifier choosing the one with the highest probability. This is a straightforward and relatively simple approach when dealing with unbalanced classes that can give us good results.

For the evaluation, **accuracy** is used as the sole measure of model performance in the paper. In unbalanced problems this is not always ideal as it can be misleading in terms of evaluating the model. Therefore, we will include our resulting F1-score for each class, along with the balanced accuracy. F1-score is generally considered a better measure of model performance in unbalanced classes situations, as it essentially takes into account both false positives and negatives. It is defined as an average between the **precison** and **recall** as such:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{1}$$

In addition, a low F1 score indicates a misclassification of the minority class, which we want to avoid in our classifier.

3 Improving the Classification

3.1 Neural Network

The first model we worked on is the Multi-Layer Perceptron. The paper used a rather simple architecture of one hidden layer with 100 neurons. For our model, we wanted to improve the complexity of theirs significantly (i.e. more layers and more neurons) without the model becoming overspecialized for the task, and thus overfit. Hence, once we settled on a 3 layer structure with 512 neurons each, we opted for a high dropout rate of 50% at every layer, together with batch normalization. These measures help prevent overfitting and stabilize the model's training. On top of these, we applied L2 regularization at every layer to penalize large weights, this together with batch normalization avoids overspecialization. Furthermore, we added a skip connection from the first layer to the output, to allow the model to carry the lower-level features of the data, captured by the first layer, to the output. This allows the other layers to act as refiners on these features, with the skip connection combining all of this information in the final dense layer. We show a representation of the architecture in Figure 2.

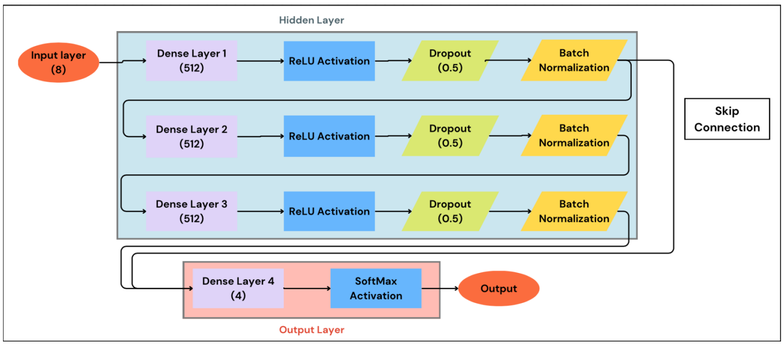


Figure 2: Diagram of neural network classifier architecture

3.1.1 Train/Test Split

Given the limited size of our dataset compared to the paper, we decided not to balance the classes by undersampling the more popular classes, as mentioned above. Instead, we kept all data and searched for the training/testing split that gives the best and most balanced results across the board. The first-hand accuracy results of running our neural network with different splits can be found in Table 1.

Table 1: Class accuracy of the classifier for different train/test split of the data. Highest overall accuracy split in bold.

Data Split (train/test)	Star-Forming	Composites	AGN	LINERs
80/20	0.880	0.729	0.701	0.736
70/30	0.904	0.719	0.726	0.716
60/40	**0.888**	**0.718**	**0.743**	**0.724**
50/50	0.899	0.676	0.721	0.741

3.1.2 Optimal Model Results

While training the model, we applied a standard scaler to normalize the inputs and make training on the data easier for our model. After training the model and classifying the test set, we obtain the confusion matrix in Figure 3.

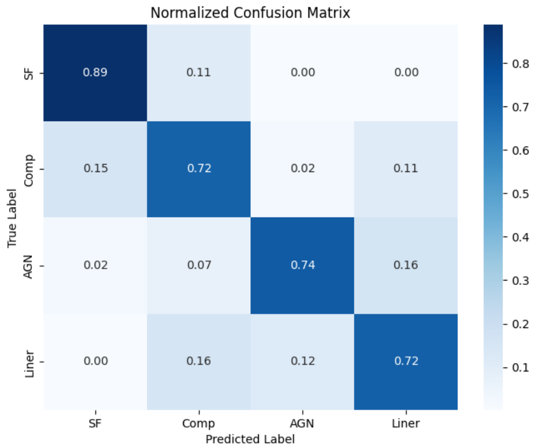
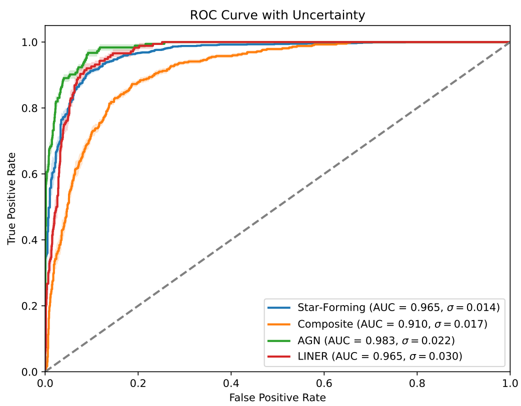
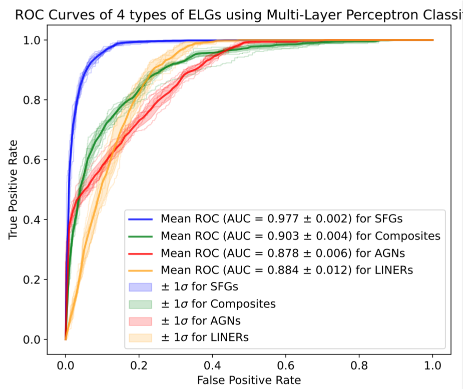


Figure 3: Confusion Matrix of our MLP Classifier

Our model results in a balanced accuracy of $\approx 77\%$, a great result for such an unbalanced classes problem. The weighted accuracy is even higher, at 82%. It is important to mention that these results are achieved thanks to the threshold adjustments applied to the classifier. Without these thresholds, the results are considerably worse and we do not outperform the paper in that case. The exact accuracy per class and F1 scores are reported in Table 2.



(a)



(b)

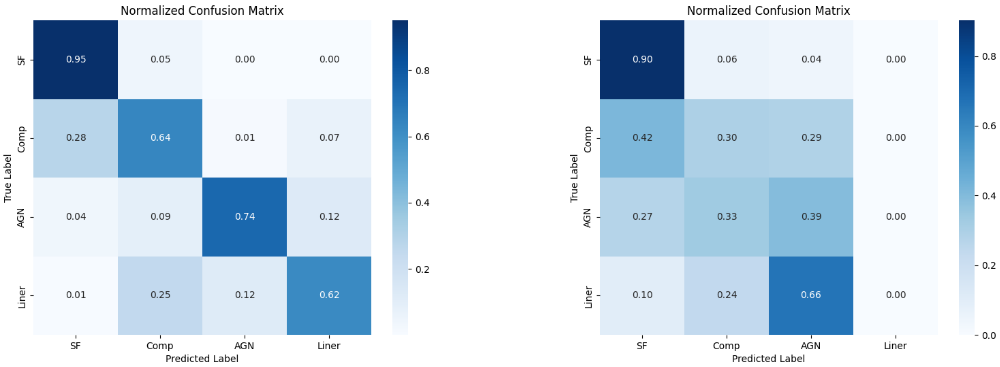
Figure 4: (a) Our neural network ROC curves / (b) Multi-Layer Perceptron results recreated from paper's code

In Figures 4a and 4b we show the ROC curves of our classifier (a) and the paper's (b). The AUC results for the underrepresented classes of AGN and LINERs are considerably higher for our model, composites are comparable, while for star-forming ours are slightly worse. This trade-off is more than favourable as our results are more balanced across the board and above 90% for all classes.

The uncertainty reported is the systematic uncertainty of our classifier calculated using the Monte Carlo Dropout method. This method entails classifying each object multiple times (in our case 50), with part of the model turned off every time and extrapolating the standard deviation from the results. Our model's architecture is designed to be quite robust with no overspecialized neurons, hence the resulting uncertainty is relatively low.

3.1.3 Neural Network: Only Optical Features and Only Colors

In the paper, the authors claim that a model that only considers the optical features of the data performs similar to the full model. We tested this by running our model on the optical features only (4) and further by running a model only on the colors (4). We show our results in Figure 5.



(a) (b)

Figure 5: (a) Confusion matrix of MLP classifier using only 4 optical features. With balanced accuracy: 0.74. (b) Confusion matrix of MLP classifier using only 4 color features. Balanced accuracy: 0.40

Unsurprisingly, the model that only trained on the color features of the data greatly underperforms and is unable to correctly classify the galaxies. The major class obtains a high accuracy purely because it is the larger class, representing almost 60% of the data. It is worth mentioning that no galaxy is found to be a LINER, while 66% of LINERs are classified as AGNs. This indicates that from only color features the model cannot tell them apart and classifies them as AGNs as there are more of those in the data than LINERs.

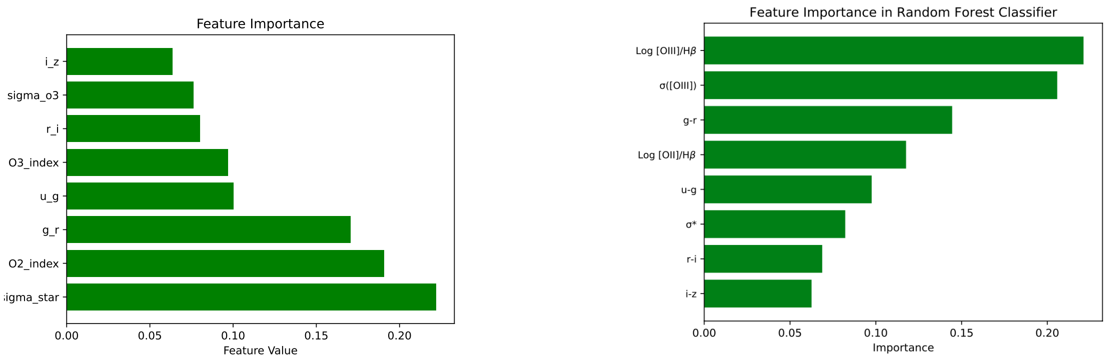
On the other hand, corroborating the paper's results, the model trained on optical features performs almost as well as the full-fledged model. It attains a balanced accuracy of 74%, only 3% lower than the full model. This indicates that the optical features are the most important ones in distinguishing these galaxies, especially among the underrepresented classes (Composites, AGNs and LINERs).

3.2 Random Forest

The second model we aimed to improve is the Random Forest classifier. According to the paper this model reached the best performance in classifying the galaxies in this dataset. Our aim is to recreate their best results and, if possible, marginally improve them. A random forest is a collection of decision trees, where each tree works on a random subset of the data. Based on its own criteria, each tree casts a vote on what the object should be and the majority "wins". For our forest we used the sklearn package '*RandomForestClassifier*', using 1000 decision trees with maximum depth of 20 and Gini entropy as purity measure. We report here the variables we ultimately settled on that obtain the best performance:

```
rf_classifier = RandomForestClassifier(n_estimators=1000,
criterion='gini', max_depth=20, min_samples_split=2,
min_samples_leaf=1, min_weight_fraction_leaf=0.0,
max_features='sqrt', max_leaf_nodes=None,
min_impurity_decrease=0.0, bootstrap=True, oob_score=True,
n_jobs=-1, random_state=None, verbose=0,
warm_start=False, class_weight = "balanced_subsample",
ccp_alpha=0.0, max_samples=None, monotonic_cst=None)
```

Classifiers such as a Random Forest allow us to determine the feature importance in a straightforward manner. Based on our MLPs trained on limited features we already expect optical features to "dominate". The resulting feature importance ranking can be found in Figure 6a, along with the paper's in Figure 6b.



(a) **(b)**

Figure 6: (a) Feature ranked by importance in our RF / (b) Paper's ranked features in RF

According to our forest, the optical features velocity dispersion and oxygen's line width are the most important features, followed closely by the g-r color band. It is worth pointing out that our ranking does not match the one obtained by the paper. We attribute this to the fact that we train on roughly a fifth of the galaxies they train on, and our classes are not balanced. This means that in our case the more important features for the majority class probably dominate over the features that would otherwise become important in a balanced situation, such as is the case in the paper. This is not a problem, as feature importance does not have a 'ground truth' but depends on the data structure/quality (as mentioned) and the model used (with its parameters).

The results of our classifier can be found in Figure 7 and Figure 8a. The resulting accuracies are comparable to the MLP performance, with the same resulting balanced accuracy of 77%. We applied threshold adjustments for this classifier as well, since it dramatically improved results without loss of information and/or generality. This result is also 2% higher than the paper's. Our AUC scores, compared to the paper's in Figure 8b also reflect this, with star-forming and composites being lower than theirs while AGN and LINERs improved considerably. We have thus managed to recreate the paper's results and marginally improve the balanced accuracy with our random forest classifier. The exact results can be found in Table 2.

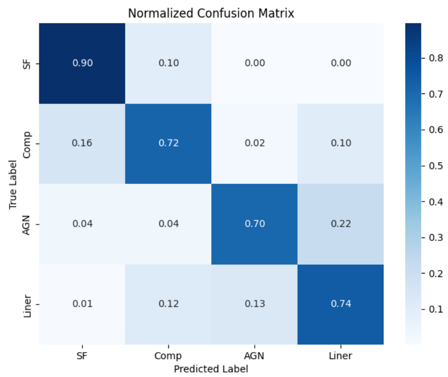
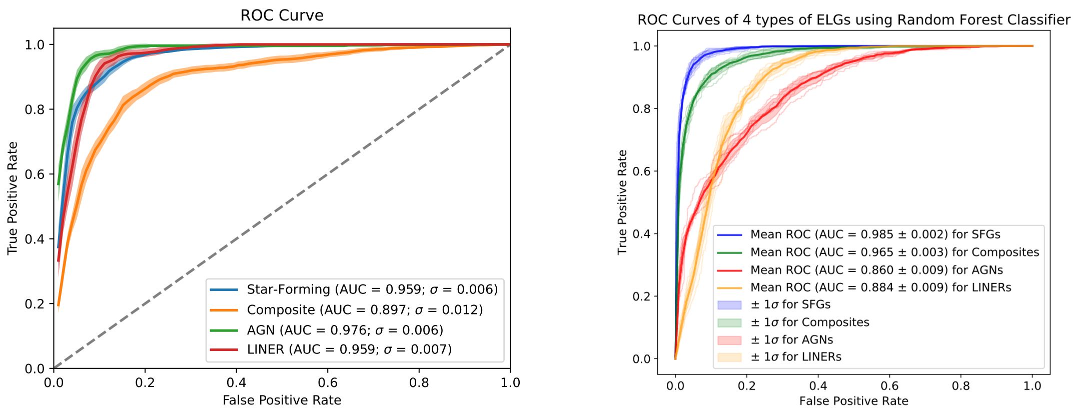


Figure 7: Confusion matrix of our Random Forest classifier

The uncertainties in our ROC curves were calculated using the bootstrap method. This entails running the model a further 25 times, retraining the model on a random subset of the data and classifying the testing set. From the True Positive Rate results we then determined the mean and standard deviation to find the uncertainties. This process helps us visualizing how predictions vary across different data sets, basically acting as an indicator of the sensitivity of our model to new data.

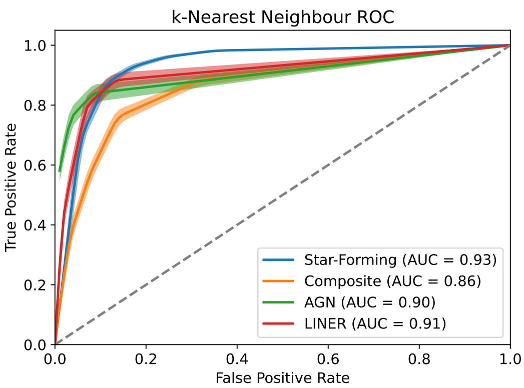


(a) (b)

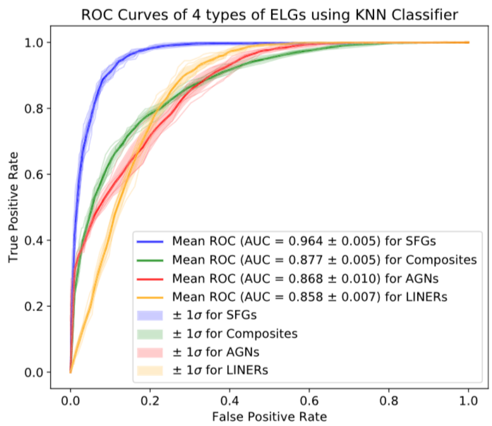
Figure 8: (a) Random Forest ROC curves / (b) Paper's Random Forest ROC curves

3.3 *k*-Nearest Neighbor and Support Vector Classifier

We ran a simple *k*-NN and SVC model on the data too. Since the paper found these to give the worst results out of the four models, we did not try and tune the models too much to give really good results, we aimed to recreate their results approximately. We show the ROC curves that we obtained for both these models together with the ROC curves of the paper in Figure 9 and Figure 10. For some classes, the AUC that we achieved is higher, while for others it is lower than the results of the paper. This is likely because we trained with unbalanced classes. Overall, the AUC is quite comparable, so we were able to reproduce the paper's results for these methods.

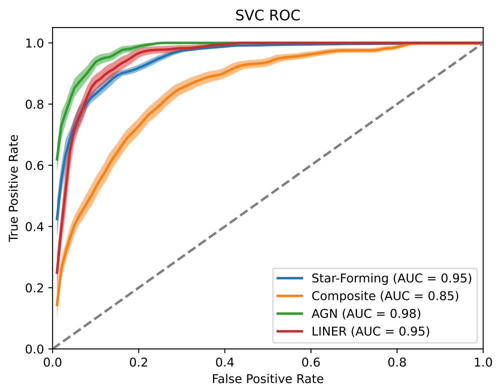


(a) Our *k*-NN ROC curves

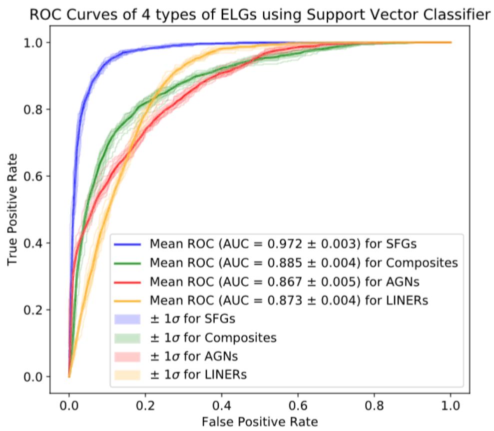


(b) The paper's *k*-NN ROC curves

Figure 9: The ROC curves for our recreation of the *k*-NN model and the SVC model.



(a) Our SVC ROC curves



(b) The paper's SVC ROC curves

Figure 10: The ROC curves for our recreation of the *k*-NN model and the SVC model.

3.4 Gradient Boosted Model

The third model we decided to build for classifying the low redshift galaxies is a boosted decision tree. A viable approach to improve the classification is to use a gradient-boosted decision tree model, since we are dealing with a multi-class classification problem. Interestingly, in the original code (*scikit_kfold_training.py*, line 443) we find as a comment:

```
# Fast but bad
clf = AdaBoostClassifier() #OneVsRestClassifier(AdaBoostClassifier())
title = 'Ada Boost Classifier'
clf = XGBClassifier()
title = 'XGBoost Classification'
```

Apparently, gradient-boosted models such as AdaBoost and XGBoost have been tested as an approach. AdaBoost was described as "fast but bad" and the XGBClassifier has no further comment as to why it was not used. However, this shall not scare us from trying to implement the XGBClassifier ourselves.

A gradient-boosted decision tree model is an ensemble method consisting of multiple weak learners which collectively improve the prediction of the model (Hastie et al., 2013). These weak learners are simple decision trees, often with little complexity. In contrast to the random forest model, gradient-boosted systems generate each new subtree to predict the residual of the ensemble created thus far. The final result is aggregated by all subtrees, however the influence of each tree is justifiably scaled based on its residual error and includes a learning rate parameter. A boosted decision tree model usually outperforms a random forest, yet for our problem we will see that they perform equally well.

It is important to note that because a gradient-boosted model builds on its previous mistakes in predicting the training data, it can quickly run into overfitting. Small changes in the training data can drastically change the structure of the trees.

One example of a gradient-boosted model implementation in python is the XGBoost library ("Extreme Gradient-Boosting"). In addition to being computationally efficient, XGBoost includes built-in regularization, which helps to reduce overfitting (for more details, see Chen and Guestrin (2016)). This makes XGBoost the obvious choice for our gradient-boosted decision tree model.

From the python *XGBoost* library, we used the `xgboost.xgbclassifier()` class, which is based on the *scikit-learn* API. We included all eight input features, as mentioned in section 2. As before, our labels are one-hot encoded so that the model does not assume a correlation between the classes. As a first benchmark, we left the hyperparameters unchanged and used these first-hand results to optimize a selection of them, with the goal of achieving better results. We also wanted to have some indication of the uncertainty of our results, so to include this in the ROC we created multiple bootstrapped versions of our test data with replacement, meaning the same source can appear twice. As in the random forest model, we then formed predictions for each resampled test data set and used the average as our prediction and the standard deviation as our uncertainty. To create a ROC, we used the function `xgbclassifier.predict_prob()` so that we have probability values for each class as an output instead of the discrete label. With this, the ROC can use different thresholds, to classify a result as positive or negative and then compare it to the true label (e.g. a threshold of 0.6 for the AGN ROC $p(\text{AGN}) \geq 0.6 \implies$ Source is an AGN).

In Figure 11, we can see that without changing anything, the model already performed very well, proving that gradient-boosted models are indeed well suited for the task. The results are also in line with our other approaches. There are plenty of parameters to adjust and fine-tune, but we focused on the following four:

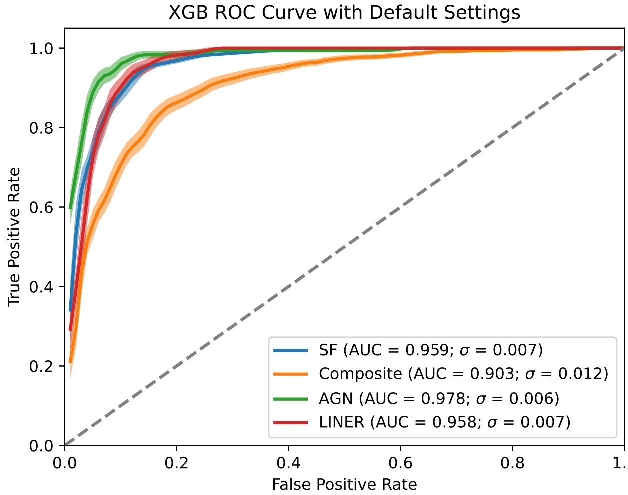


Figure 11: The ROC curve obtained using the default XGBClassifier. The default values of the parameters we will modify later are $d = 3$; $n_e = 100$; $\eta = 0.3$; $\lambda = 1$

1. `max_depth` (d): the maximum depth each tree is allowed to generate, meaning how many decision layers there are. Its default setting is $d = 3$.

2. `n_estimators` (n_e): The number of subtrees that are generated in the iterative process. If it is left unchanged, the model will create 100 decision trees.

3. `learning_rate` (η): similar to other machine learning models, we can influence how much the system is affected by the previous residuals. A smaller learning rate can reduce overfitting by slowing down the adaption of each tree to the ensemble. However, if n_e is set too low, the process might stop too early. The default value is $\eta = 0.3$.

4. `reg_lambda` (λ): determines how much regularization takes place. If $\lambda = 0$, we have no regularization. Increasing the value leads to smaller weight updates, meaning smaller changes in the leaf nodes. When not specified, the model will use $\lambda = 1$.

To find the optimal values, we simply tested different values for each parameter and observed where the average AUC of the ROC is the highest. To optimize the process, we tuned each parameter while leaving the others unchanged. This might leave out some combinations of parameters which could possibly fully maximize the model's accuracy, but for the scope of this project it gave us solid values to work with.

Looking at the results in Figure 12, we find that the optimal amount of subtrees is 30, compared to the default of 100. This means that after 30 subtrees, any new decision tree does not improve the result and leads to overfitting, thus worsening the accuracy. We can also see that the subtrees only need a depth of 3 decision layers to be effective. It is interesting to note that given the levels of complexity possible with such models, a model with only 30 trees of little complexity can offer good results.

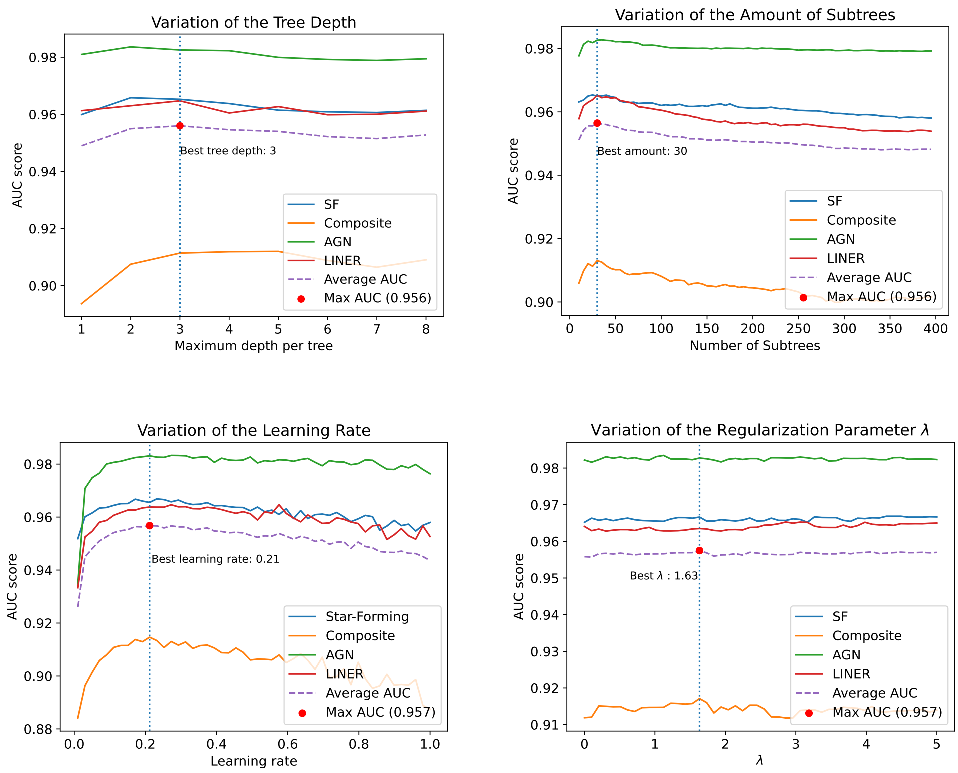


Figure 12: The change in the AUC scores for the four classes as a function of the parameters for the XGBClassifiers.

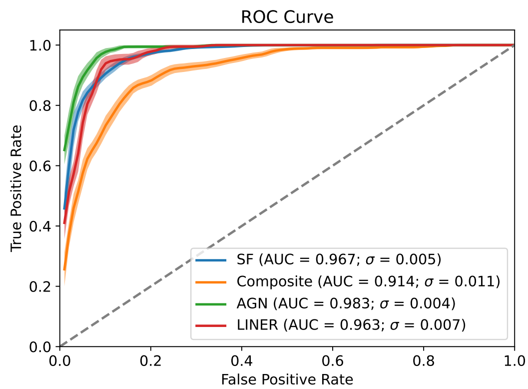


Figure 13: The updated XGBClassifier, the used parameters are now
$d = 3;\ n_e = 30;\ \eta = 0.21;\ \lambda = 1.63$

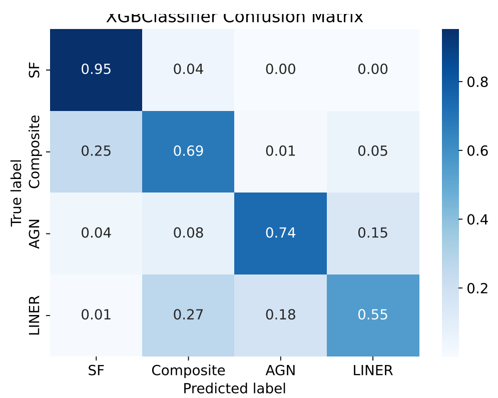


Figure 14: XGBClassifier confusion matrix

Using the XGBClassifier with the optimal parameters offers slight improvements (Figure 13). We see an improved AUC for identifying star-forming galaxies and composite galaxies, as well as a slightly reduced uncertainty for all classes except the LINER. We also find in the confusion matrix (Figure 14) that the LINER are by far the worst performing class in the XGB classification. It should be noted that we have not yet applied the adjusted thresholds, as we did in our other models, and are thus so far working on an unbalanced problem with no adjustments.

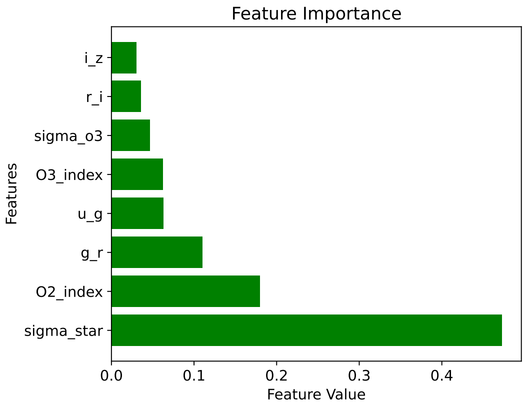


Figure 15: The importane of each feature for the XGBClassifier.

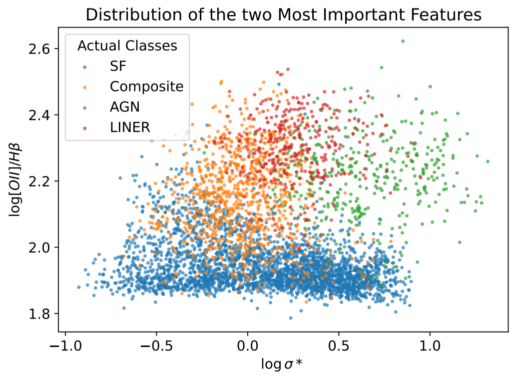


Figure 16: Feature space of the two most important features with the true training labels.

Before adjusting the thresholds, we will take a look at our feature importance in Figure 15. We can see that we have nearly the same feature importance distribution as in our other decision tree mode, the random forest. For the XGBClassifier, the stellar velocity dispersion σ_* and the OII line are the most important features for the classification. If we plot the values of just those two features (Figure 16), we can see why the XGBClassifier has troubles with identifying LINERs. They lie more or less right in the middle of the composite galaxy and AGN values. Additionally, we can see the already discussed class imbalance again, which adds difficulty to the classification. If we change the probability thresholds of the classifier to favor the underrepresented classes, we can see in Figure 17 that our LINER accuracy is much better. Of course, this comes at the cost of correctly classifying star forming galaxies, since this threshold is set much higher than the others. The balanced accuracy of the model is improved by 3% and is now comparable to our two other models.

If we compare our results with the other two models (see Table 2), we can see that we have worse accuracies, for the Composites and AGN classes, but better AUC scores. As already mentioned, we are dealing with imbalanced data sets, so accuracies aswell as AUC scores can be misleading. We thus refer to the F1 scores and the balanced accuracy as the proper measures of model performance. Furthermore, the XGB-Classifier needs considerably less computational power to get these results, which can be advantageous when performing several different tests on the data. Or, to put it briefly, we can see that a gradient-boosted model does indeed perform well while being computationally lighter than the other classifiers. This meets our expectations, and we are left wondering why this approach was omitted in the original paper.

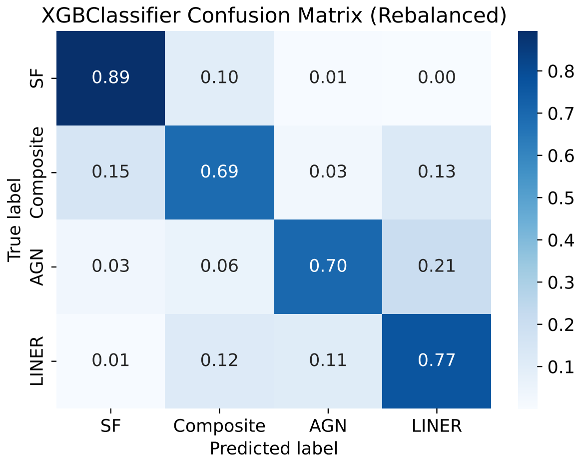
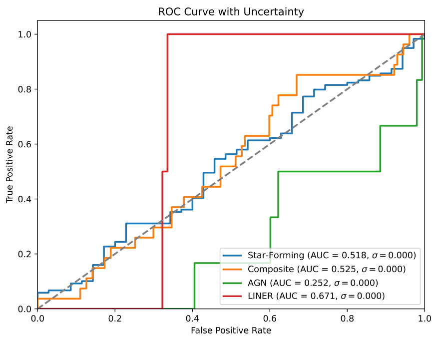


Figure 17: The re-balancing uses the adjusted thresholds of 0.7 for SF, 0.3 for composite, 0.25 for AGN and 0.3 for LINER.

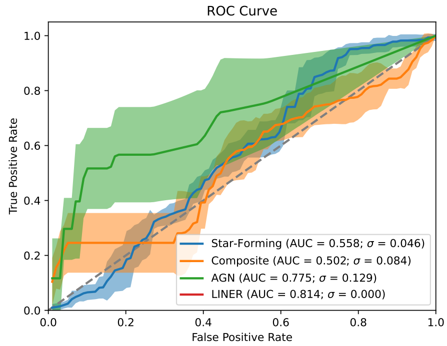
4 Testing on Intermediate Redshift

As mentioned in section 2, we wanted to test if our models can actually classify intermediate redshift sources. For this we had to obtain different data ourselves, since the data described in Zhang et al. (2019) has not yet been classified. Our intermediate redshift observations ($0.32 < z < 0.8$) also come from the SDSS survey and have already been classified by Kewley et al. (2001); Schawinski et al. (2007); Kauffmann et al. (2003b) into the four categories (what they call "Seyfert" are AGN). We filtered this data in the same way as they did in the paper, using the same eight features. Unfortunately, this left us with only 154 galaxies. Trying to predict the new data with our models did not offer very promising results. Unfortunately, none of them are able to make useful predictions. We show the ROC curves of the neural network, the random forest and the XGBClassifier in Figure 18. Most are just random guesses, except when predicting the AGN. Somehow, the XGBClassifier does particularly well on them, and the random forest also performs decent here compared to the other classes. Even our neural network has a worse-than-random AUC for the AGN class, which means it at least recognizes some pattern in the features, even if it is the wrong way around.

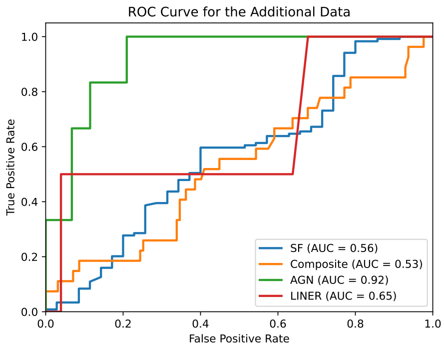
This leaves us with the question of why the results of this classification are so bad. The first reason could be the relatively small data set, but it should still show at least some indication of correct predictions, especially because we do not use this data to train on, only to classify. Then there could be a problem with our data preparation. To test this, we would need more data on intermediate redshift sources. We could then train the models on this data and see how well they perform. Poor performance would mean that there is something fundamentally wrong in the new data we have collected or how we processed it. This seems unlikely, however, since we followed precisely what was done for the training data. It is more likely that the problem lies somewhere deeper. Our current hypothesis is that the eight features that were selected in Zhang et al. (2019) may work well for predicting sources at lower redshift, but might have different relations when the redshift increases. Again, more intermediate redshift data would provide more insight here, as we could train the models on the higher redshifted data and then try to let them predict our current training data. Another likely explanation is that the data inherently differs due to some reason such as what instruments it was obtained with or the way in which the lines were fit. All of this



(a) Neural Network



(b) Random Forest



(c) XGBClassifier

Figure 18: ROC curves of our models applied on the new data.

could be explored in the future, as new data becomes available. For now, we are left in the unknown and can only guess.

5 Discussion

We summarized our results together with those of the paper in Table 2, Table 3, and Table 4.

We managed to reproduce all results from the paper, even though we trained on a much smaller and imbalanced dataset. We improved on their results with all three of our methods: the Neural Network; the Random Forest; and the XGBClassifier. We obtained slightly higher balanced AUC scores and accuracies for all three of these compared to the paper's best result, i.e. their random forest. Most importantly, we used a different method from the paper to deal with the unbalanced dataset. While the paper could afford to cut down on their data to balance out each class, we would have been left with much too little data if we were to follow their method (as mentioned in Section 2.3). Therefore we had to be creative and devise a way to deal with this imbalanced data to still achieve good results for all of the classes. This is why we decided to adjust the probability thresholds for the classification of the different classes, where we mostly dropped the threshold for the minority classes. This way of dealing with the imbalanced data ended up giving us better results than the paper's. We cannot know how our models would behave if we trained on the same dataset as the paper did, but we would expect our models to perform even better, since more training data usually gives better results for such classifiers. For future analysis it could be really interesting to try out our method on the entire dataset that the paper used, or inversely, to apply SMOTE or other oversampling techniques to our dataset. The way they mention they deal with the imbalanced dataset would leave them with less than 10 000 galaxies from the over 28 000 sample, so using our method the models could be trained with almost three times more data, which could produce better results.

As mentioned in Section 2.3, the paper also has some shortcomings that could be improved upon. Their way of classifying the galaxies could be problematic, as they do not use all possible optical lines (i.e. they omit [O I]) and do not use all criteria from Kewley et al. (2006). This might result in some wrong labels that we feed our machine learning models which would be highly problematic for the accuracy of the models with reality. Additionally, as long as we do not have any IR data of the large dataset of intermediate redshift galaxies that they classified, we have no way of knowing if their classification of those is correct and thus whether the models actually generalize to higher redshifts. Finally, as they mention themselves too, and as we found out with the extra data we obtained, the models are not generalizable to other datasets and thus cannot help with classifying other datasets, nor can their performance be verified using other datasets.

Our results might also be improved with time since machine learning methods change and advance rapidly. Additionally, once IR data is available for the intermediate redshift galaxies, besides verifying the performance of the models on these, they could be trained on this data too to make the model more precise and maybe even allow for higher redshift galaxies to be classified. Additionally, combining one large dataset from different surveys and instruments might be interesting too, to see if good performance can be achieved on different data.

Table 2: Accuracy, F1 and AUC scores of our models in classifying low redshift galaxies. 'Overall' is the weighted average over all classes.

	Neural Network			Random Forest		
	Accuracy	F1-score	AUC	Accuracy	F1-score	AUC
Star Forming	0.889	0.910	0.965 ± 0.014	0.897	0.920	0.959 ± 0.006
Composite	0.718	0.680	0.910 ± 0.017	0.717	0.700	0.897 ± 0.012
AGN	0.743	0.770	0.983 ± 0.022	0.705	0.740	0.976 ± 0.006
LINER	0.724	0.670	0.965 ± 0.030	0.744	0.670	0.959 ± 0.007
Overall	0.825	//	//	0.826	//	//
Balanced	0.768	0.760	0.956	0.766	0.758	0.948

		XGBClassifier	
	Accuracy	F1-score	AUC
Star Forming	0.894	0.910	0.967 ± 0.005
Composite	0.689	0.670	0.914 ± 0.011
AGN	0.699	0.730	0.983 ± 0.003
LINER	0.767	0.670	0.963 ± 0.007
Overall	0.820	//	//
Balanced	0.762	0.745	0.957

Table 3: The AUC scores of the different classifiers from Zhang et al. (2019) for the four different categories and the overall AUC.

Classifier	KNN	SVC	MLP	RF
Star-forming Galaxies	0.964 ± 0.004	0.968 ± 0.005	0.973 ± 0.003	0.985 ± 0.001
Composites	0.878 ± 0.010	0.881 ± 0.014	0.896 ± 0.009	0.966 ± 0.004
AGNs	0.860 ± 0.007	0.861 ± 0.009	0.880 ± 0.004	0.876 ± 0.008
LINERs	0.865 ± 0.008	0.869 ± 0.009	0.877 ± 0.009	0.897 ± 0.004
Balanced	0.892	0.895	0.906	0.931

Table 4: The accuracies of the different machine learning algorithms for the four different categories and overall.

Classifier	KNN	SVC	MLP	RF
Star-forming Galaxies	0.934 ± 0.001	0.926 ± 0.003	0.937 ± 0.004	0.934 ± 0.001
Composites	0.596 ± 0.006	0.638 ± 0.008	0.688 ± 0.017	0.694 ± 0.005
AGNs	0.760 ± 0.008	0.794 ± 0.004	0.768 ± 0.022	0.718 ± 0.011
LINERs	0.516 ± 0.009	0.608 ± 0.007	0.611 ± 0.019	0.657 ± 0.010
Balanced	0.702	0.741	0.750	0.751

6 Conclusion

For this project, we have recreated and improved on the results of Zhang et al. (2019). The main aim was to train different machine learning models on low redshift (< 0.3) emission-line data of galaxies to classify them into one of four categories based on their main ionization source: star forming, active galactic nuclei, composites, and low-ionization nuclear emission regions. We recreated all four models the paper used and we improved on their results with a neural network, a random forest, and a boosted decision tree model. We managed to slightly improve on their best model, their random forest, even though we trained on less data and on an imbalanced dataset. We dropped the threshold for the classification of the underrepresented classes in order to deal with the imbalanced dataset and this is how we achieved slightly higher accuracies and AUCs for each of our models regardless of the imbalanced data.

7 Contributions

The separation of the project's workload has been equal throughout the project. Csilla worked on transcribing the paper's code and making sure it worked correctly by recreating their results, she also re-wrote parts of it so that it was ready and straightforward to use. Lorenzo designed, coded and optimized the Neural Network and Random Forest classifiers. Felix designed, coded and optimized the boosted decision tree classifier. The separation in these tasks is not as clear cut as we all cooperated in almost every task and helped each other. Furthermore, Csilla worked on obtaining the galaxy data at intermediate redshifts from astronomical data surveys. Finally each of us participated equally in writing the report, contributing mostly in the sections regarding the subject we worked on while coding and proof-reading other's sections for eventual fixes. Finally, Felix mostly organized the GitHub (README file, etc.).

This contributions section was written with the approval of all group members: Lorenzo Cesario, Felix Schulz and Csilla Tijssen.

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